Birds Classification based on Bird Sound

by

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Dissertation submitted to the Faculty of the Engineering of the
Savitribai Phule University of Pune, in partial fulfillment
of the requirements for the degree of
Masters in Electronics and Telecommunications
Year

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CERTIFICATE

This is to certify that the thesis titled **"Birds Classification based on Bird Sound"** submitted by **Girish Lumdeo Rane (121997014)** in partial fulfillment of the requirements for the award of Master of Technology degree in Electronics and Telecommunication Engineering, with specialization in **Signal Processing** during year 2020-21 at College of Engineering, Pune is an authenticate work by her under my supervision and guidance.

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Dissertation Approval

This dissertation entitled

Birds Classification based on Bird Sound

by
Girish Lumdeo Rane

is approved for the degree of

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College of Engineering Pune
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Examiners	Name	Signature
1. External Examiner	________________	________________
2. Guide/Supervisor	________________	________________

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Acknowledgment

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The Mtech project experience with above all have indulged in me important values and principles which are going to help me throughout my life.

(Girish Lumdeo Rane)

ABSTRACT

Automatic classification of bird calls plays an essential role in monitoring and protecting biodiversity. Classifying bird species is a challenging problem due to variant species present in the world. This work is focused on identifying different bird species. This project deals with various feature extraction techniques to classify species of birds. Also, feature selection is carried out to reduce the complexity and increase the accuracy of classifiers. The MFCCs, LFCCs and GFCCs have been extracted directly from birds recording. Also, the methodology is tested on four different classifiers viz K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Artificial neural networks (ANNs) and Convolutional Neural Network(CNN).

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List of Abbreviations

FC	Fully Connected
DCT	Discrete Cosine Transform
KNN	k-Nearest Neighbors
STFT	Short Time Fourier Transform
MFCC	Mel Frequency Cepstral Coefficient
LFCC	Linear Frequency Cepstral Coefficient
GFCC	Gammatone Frequency Cepstral Coefficient
SVM	Support Vector Machine
ANN	Artificial Neural Network
CNN	Convolutional Neural Network

Chapter 1

Introduction

1.1 Motivation

Last few decades, lots of research is carried out in the analysis of bird audio signals. Birds are good indicators of the health of the environment. They play a significant role in the ecosystem. Identification of bird species can be made through image, video and audio. Audio processing is widely used for capturing species. The greatest challenge is identifying bird species of interest within large recordings. Therefore, the auto- matic approach is used for the classification of species. Various techniques are used to identify acoustic, such as Mel Frequency Cepstral Coefficients (MFCCs), Linear Frequency Cepstral Coefficients (LFCCs), which summarize spectral information. This project focused on the identification of bird species from their sound. Each bird has a different tone and pitch. The aim is to identify each bird by its sound by removing environmental noise. The deep learning approach is used for the classification of species into different categories.

1.2 Objectives

1. To create a dataset of bird sound.
2. To study and Extract features from bird sound.
3. To select optimum feature for bird classification.
4. To perform classification of bird species using different machine learning algorithms.
5. To compare the performance of various classification algorithms.

1.3 Organization of Thesis

Chapter 1 provides an introduction to the project.

Chapter 2 gives the literature survey of various approaches.

Chapter 3 presents different Feature Extraction and Selection Techniques.

Chapter 4 explain different classification methods.

Chapter 5 discuss experimental data and result of classifiers

Chapter 6 provides the conclusion and future scope.

1.4 Summary

This chapter introduces the fundamental idea of this thesis. It gives the motivation behind automatic bird classification. It covered a preview of how the thesis is planned and gives the whole concept of the project.

Chapter 2

Literature Review

2.1 Introduction

The deep learning model has recently attracted a lot of attention due to its high performance in building automatic bird sound classification systems. Support Vector Machine(SVM), k-nearest neighbour (KNN), Artificial Neural Network(ANN) are used as classifiers. Literature of various approaches that are widely used in audio classification is explored for the recognition of bird sounds.

2.2 Birds Classification using various methods

Charisma, M. R. Hidayat and Y. B. Zainal, showed the use of Mel Frequency Cepstral Coefficients with Vector Quantization for speaker verification. VQ is used to produce a vector of each speech signal. Vector quantization creates a cluster of each vector area called codebook.

P. Jancovic, M. K ˇ ok¨ uer, M. Zakeri, and M. Russell suggested tonal-based features to identify tonal bird sounds. The Gaussian mixture model is used to model 95 bird species. Each bird sound contains white noise and environmental noise.The Tonal-based approach provides better performance compared to Mel Frequency Cepstral Coefficients.

Lee, Chang-Hsing Lee, Yeuan-kuen Huang, Ren-zhuang suggested a new method to identify species. First, average MFCC and average LPCC are calculated to create a feature vector. Then, linear discriminant analysis is used to increase the accuracy of the feature vector.

Trifa VM, Kirschel AN, Taylor CE, Vallejo EE have used the Hidden Markov Model for acoustic bird classification. The work uses five bird species without noisy recording.

Dewi, Sita & Prasasti, Anggunmeka & Irawan, Budhi designed a new feature extraction technique called Linear Frequency Cepstral Coefficients (LFCC). LFCC uses a Linear Filter bank K-nearest neighbour algorithm is implemented to detect baby is crying or not.

2.3 Birds Classification using classifier

J. Xie, K. Hu, M. Zhu, J. Yu and Q. Zhu, proposed different types of time-frequency representations called Mel, harmonic and percussive spectrogram. CNN fused model gives balanced accuracy of 86.31 %.

Á. Incze, H. Jancsó, Z. Szilágyi, A. Farkas and C. Sulyok, , have developed a methodology to classify species using spectrogram. A pre-trained MobileNet CNN model was used to differentiate species. STFT of each audio file given to input to CNN model. This experiment shows the colour map of the spectrogram gives more advantages than a normal spectrogram.

Patole R., Rege P. suggested identifying birds by their sounds using Mel Frequency Cepstral Coefficients (MFCC) and Gammatone Frequency Cepstral Coefficients (GFCC) for classification. The feature set is evaluated using the Support Vector Machine and Artificial Neural Network.

2.4 Summary

In this chapter, literature related to birds classification is explored. Birds classification can be done in the traditional method and deep learning approach. Various features used by earlier researchers is also discussed

Chapter 3

Feature Extraction and Selection

3.1 Introduction

In this Chapter, Different feature extraction techniques are discussed. This technique extract features from the audio signal. Also, a procedure for the best feature selection is demonstrated. Each method discusses the steps of deriving a feature vector from bird sound.

3.2 Mel Frequency Cepstral Coefficients

Mel frequency cepstral coefficients have been used for speaker recognition. Speaker recognition is mainly divided into two parts, identification and verification. For verification, the goal is to verify the speaker. For identification, the goal is to determine which group matches the voice of the sample species. Feature extraction is the extraction of small information from voice data that represent the speaker. This information is stored in vector form and used for the classification of bird species.

The MFCCs are used in the feature extraction process. The block diagram of MFCC is shown in Figure 3.1,

The first step is Pre-Emphasis. Pre-emphasis refers to filtering that emphasizes the higher frequencies. Framing Each bird signal has framed the block with overlapping. The framing process divides the whole audio signal into small frames. The Hamming window is applied to each frame. Fast Fourier Transform converts small frame signal

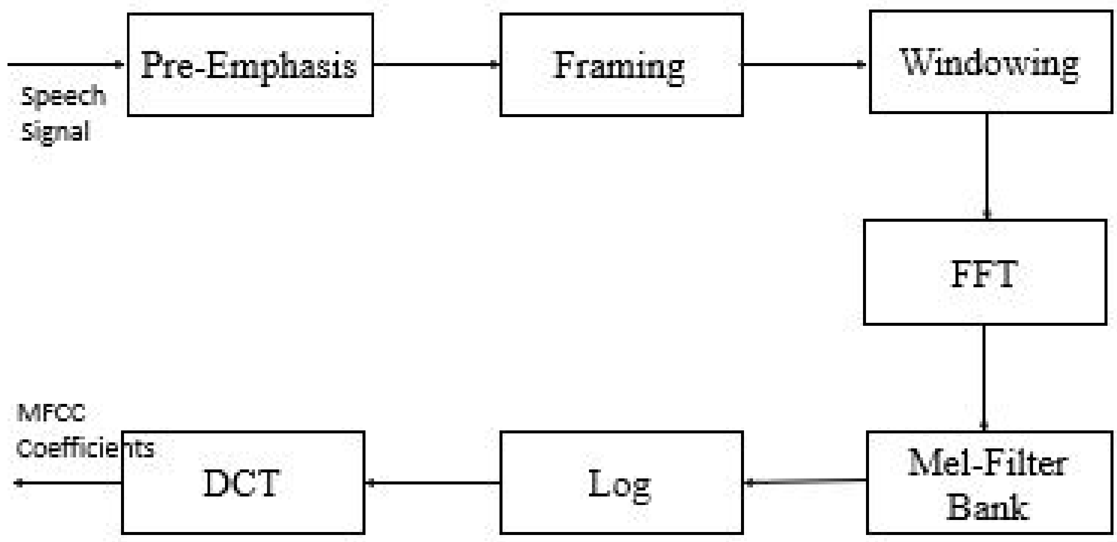


Figure 3.1: Overview of steps involved in the extraction of MFCCs

to the frequency domain. The equation to convert the normal frequency to Mel frequency is given by,

$$Mel(f) = 2595 * \log(\frac{f}{700}) \tag{3.1}$$

The calculated power spectrum is mapped with Mel frequency with the help of a filter bank. The filter bank is narrow in the lower frequency range and wider in the higher frequency range. The equation of triangular filter bank given as,

$$H_m(k) = \begin{cases} 0, & \text{f}(m-1) > k \\ \frac{k-f(m-1)}{f(m+1)-f(m)}, & f(m-1) \geq k < \text{f(m)} \\ \frac{f(m+1)-k}{f(m+1)-f(m)}, & f(m) \leq k \leq \text{f(m+1)} \\ 0, & \text{f}(m+1) < k \end{cases} \tag{3.2}$$

Where f() is the list of M +2 mel-spaced frequencies and M is the required number of filters. The design filter bank is applied on each power spectrum.The logarithm of all these summed values has been obtained and is referred to as a log mel spectrum. Discrete Cosine Transform of log scale spectrogram is MFCC coefficients.

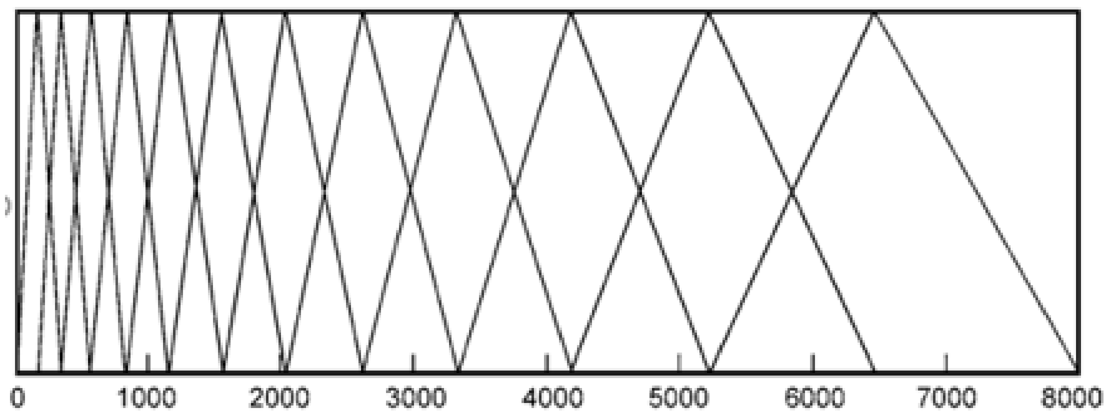


Figure 3.2: Mel Filter Bank

3.3 Linear Frequency Cepstral Coefficient

LFCC uses a Linear frequency scale. The idea behind Linear frequency is a linearly designed frequency band. The difference between the two filter banks is the same. Each filter bank is equally divided. The resolution of frequency in the Mel filter bank decreases for a higher frequency, but in linear frequency, The resolution of the filter bank is the same. A total of 32 filter banks were used. The block diagram of Linear Frequency Cepstral Coefficients is shown in Figure 3.3.

3.4 Gammatone Frequency Cepstral Coefficient

The Gammatone Frequency Cepstral Coefficients (GFCC) are auditory characteristics based on a set of Gammatone filters. The gammatone filters are designed to simulate the process of the human auditory system. The equation of centre frequency fc is given by,

$$g(t) = at^{n-1}e^{2\pi bt}\cos(2\pi f_c t + \phi) \tag{3.3}$$

Where ϕ is the phase of the filter. Usually, it is set to zero. The constant values a and n defines the order of the filter. The value of a must be less than 4.

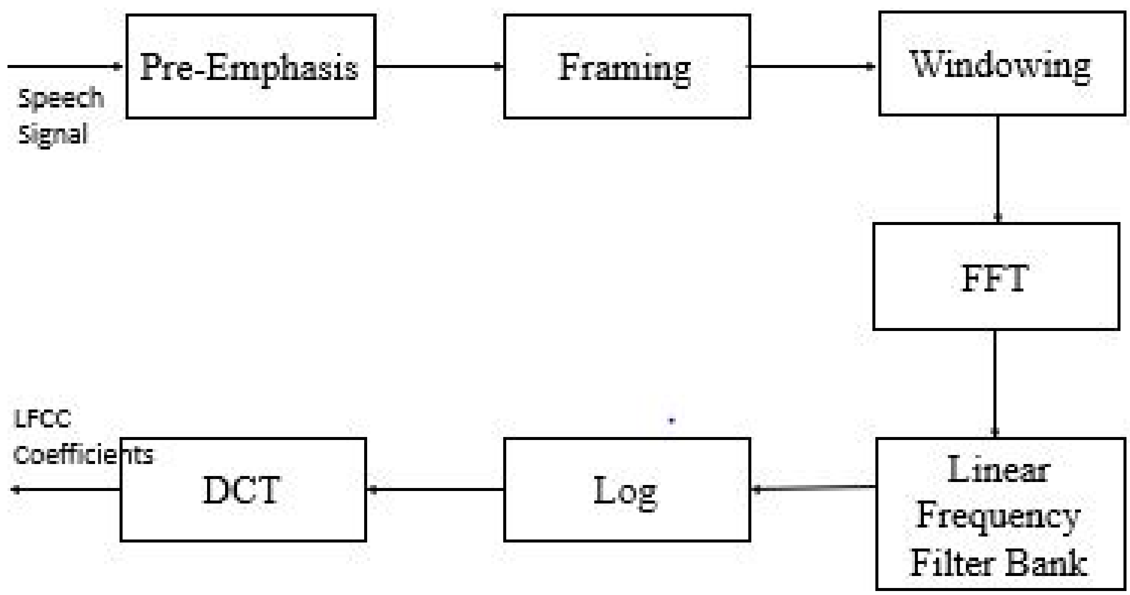


Figure 3.3: Overview of steps involved in the extraction of LFCCs

The factor value b is given as,

$$b = 25.17 * (\frac{4.37fc}{1000} + 1) \tag{3.4}$$

The block diagram of Gammatone Frequency Cepstral Coefficients is shown in figure 3.4,

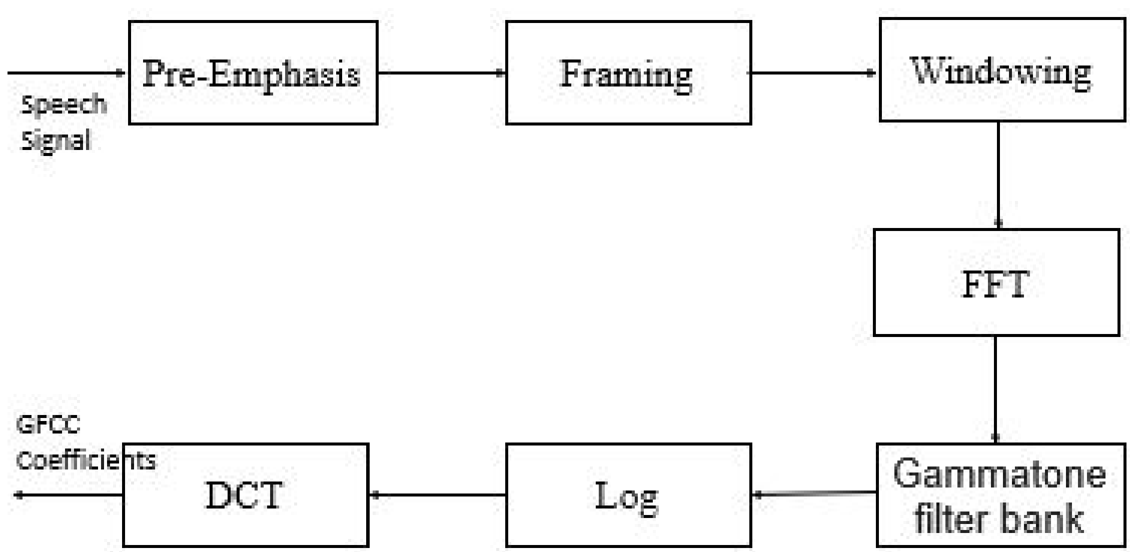


Figure 3.4: Overview of steps involved in the extraction of GFCCs

3.5 Feature Selection

Feature selection is a crucial optimization algorithm for classification. Feature selection selects only the relevant features and discards all irrelevant and noisy feature.Thus, it is an unrestrained target algorithm. It improves the efficiency and accuracy of the model. The advantages of this approach is to select a small set of features that perform well in an algorithm. The benefits of feature selection are that the learning algorithm only focuses on essential data and a more accurate model without a noisy feature.

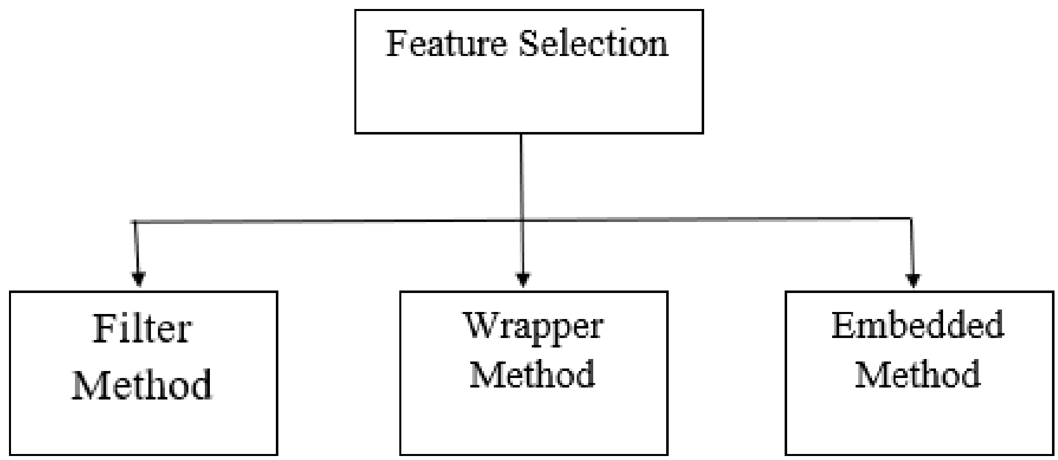


Figure 3.5: Block Diagram of Feature selection methods

The Feature selection algorithm divided into three methods called Filter method, Wrapper method and Embedded method. Filter method uses to evaluate a subset of feature and wrapper method provide the best subset of features. Figure 3.5 shows the block diagram of different feature selection method.I have used wrapper method for feature selection. most commonly used technique under wrapper method are Forward selection,Backward elimination,Bi-directional elimination and I have selected Recursive Feature Elimination(RFE) algorithm.

3.5.1 Filter Method

It is a rarely used method for feature selection. It filters out data from the characteristics of features. The main advantage of the filter method is it is computationally more accessible than the wrapper method. It assigns the ranking and subset of features from a set of all features. Here, The selection of features is independent of classifiers. It is a high correlation with target variables and a low correlation with another independent variable. In this method, features are selected based on four primary subset generation techniques viz., heuristic feature subset selection, forward selection, backward elimination, and bidirectional selection.

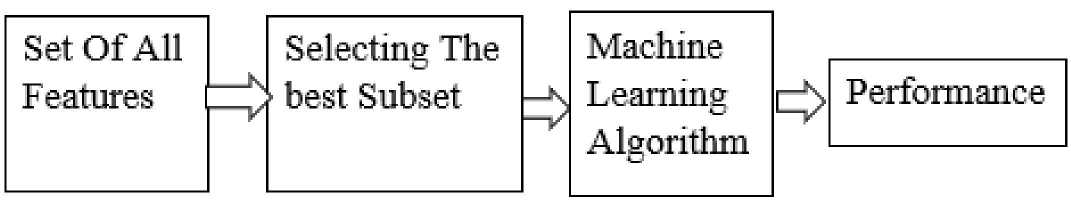


Figure 3.6: Block diagram of the filter method

3.5.2 Wrapper Method

The wrapper method is a greedy search algorithm that finds the minimum number of features by recursive approach. It also selects features based on the performance of classifiers. It starts with an empty set of features, then pools feature from all feature collections and evaluate performance in the algorithm. This process is done recursively. It is also called recursive feature selection. Examples of wrapper methods are Recursive feature elimination, genetic algorithm, and sequential feature selection algorithms. Figure 3.7 shows Block Diagram of Wrapper method.

The most commonly used feature selection is Recursive Feature Elimination(RFE). RFE generally removes the least important features and retains the remaining feature. The method for feature selection is used in this work. The most commonly used technique under wrapper method are Forward selection,Backward elimination,

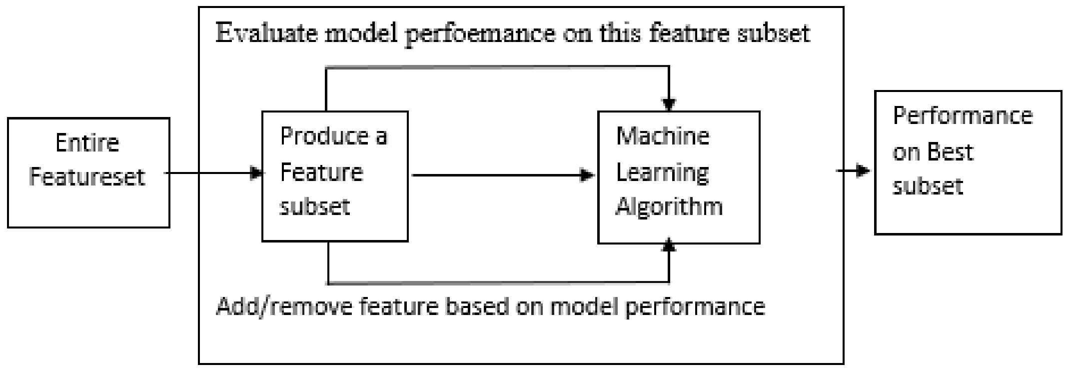


Figure 3.7: Wrapper method for feature selection

Bi-directional elimination. Recursive Feature Elimination(RFE) algorithm is explored in the work.

3.5.3 Embedded Method

The embedded method is the combination of both filter and wrapper methods. The advantage of this algorithm is more accurate and own variable selection to perform classification. This algorithm split all features into all possible ways and selects the best set of features. The most common example is Random Forest Feature Selection. The Block Diagram of the Embedded method is shown in Figure 3.8.

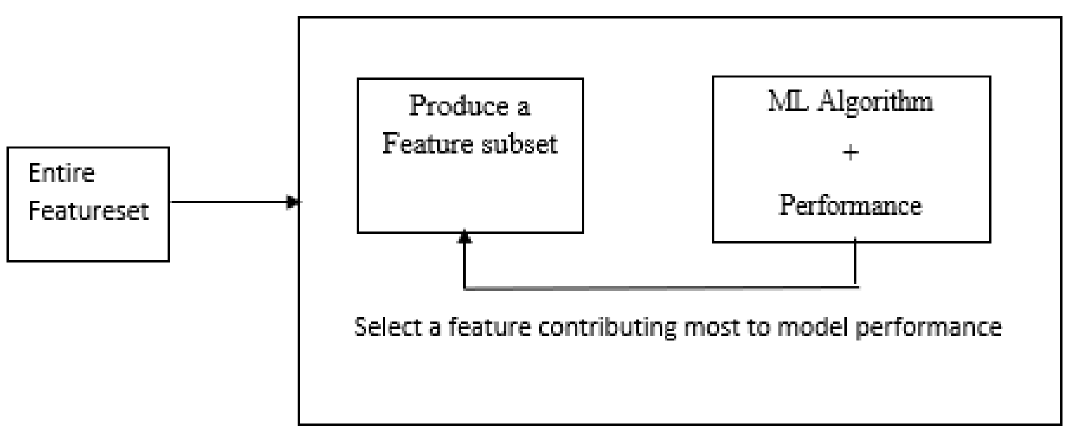


Figure 3.8: Embedded method for feature selection

3.6 Summary

In this chapter, we discuss different types of Feature extraction and Feature selection techniques. These techniques are helpful for analyzing audio signals. The most commonly used technique is MFCC, i.e., Mel Frequency Cepstral Coefficients. Another two techniques are also beneficial for the application of bird species classification.The computational complexity of the wrapper method is more than the filter and embedded method. Recursive Feature Elimination (RFE) algorithm select features recursively.

Chapter 4

Classification

4.1 Introduction

For classification, Each audio signal is represented as a feature vector. This feature vector represents the input to the classifier. K-nearest neighbors classifier is a supervised machine learning algorithm along with a Support vector machine (SVM). Artificial Neural Network (ANN) and Convolutional Neural Network (CNN) are also discussed.

4.2 K-Nearest Neighbor

It is a supervised machine learning algorithm used for regression and classification. The classification is carried out by a simple majority vote of the nearest neighbor of each query point. For each query point, the value of k is defined by the user. Each query point considers the closest k value and predicts the maximum possible class. The advantage of this algorithm is straightforward and gives promising results. It is also used for identifying similar objects. For this algorithm, there is no need to build the model or set the parameter. For each bird audio signal, the mean value of the feature is extracted to form a vector. This vector acts as input to the KNN algorithm. From experiment data, k=3 gives better accuracy than other k value.
From Figure 4.1, The test sample is either class blue or class red. For the value of k is 3, then the three nearest samples are considered and if the value of k is 5 then the five closest samples are considered.

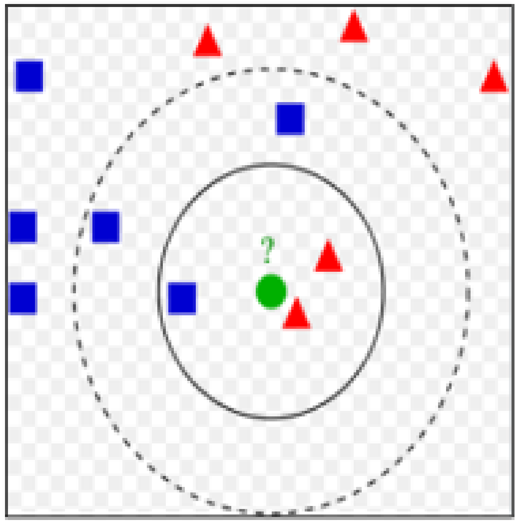


Figure 4.1: Example of KNN classifiers

4.3 Support Vector Machine

Support Vector Machine is a supervised machine learning algorithm used for classification and regression problems. The main goal of the support vector machine is to find a hyperplane between two classes. For linear and non-linear problems, support vector machines play an important role. To separate two different classes, the margin between two classes is maximum. Here, the mean value of each feature vector is calculated. This mean value gives input to the support vector machine. Then, the support vector machine calculates the marginal distance between the two classes. For classification, the linear kernel provides the highest accuracy.

4.4 Artificial Neural Network

An Artificial Neural Network is a Neural network designed the way the human brain simulates. ANN also consists of neurons that arrange in layers. Generally, ANN contains three layers called the Input layer, hidden layer and output layer. The hidden layer has more than one layer. Each layer has two or more neurons. The ANN uses a training algorithm that learns and updates weights by the backpropagation algorithm. The ReLU (Rectified Linear Unit) activation function assigns value zero if the negative and same value is positive. The block diagram of the ANN algorithm is shown in fig 4.2

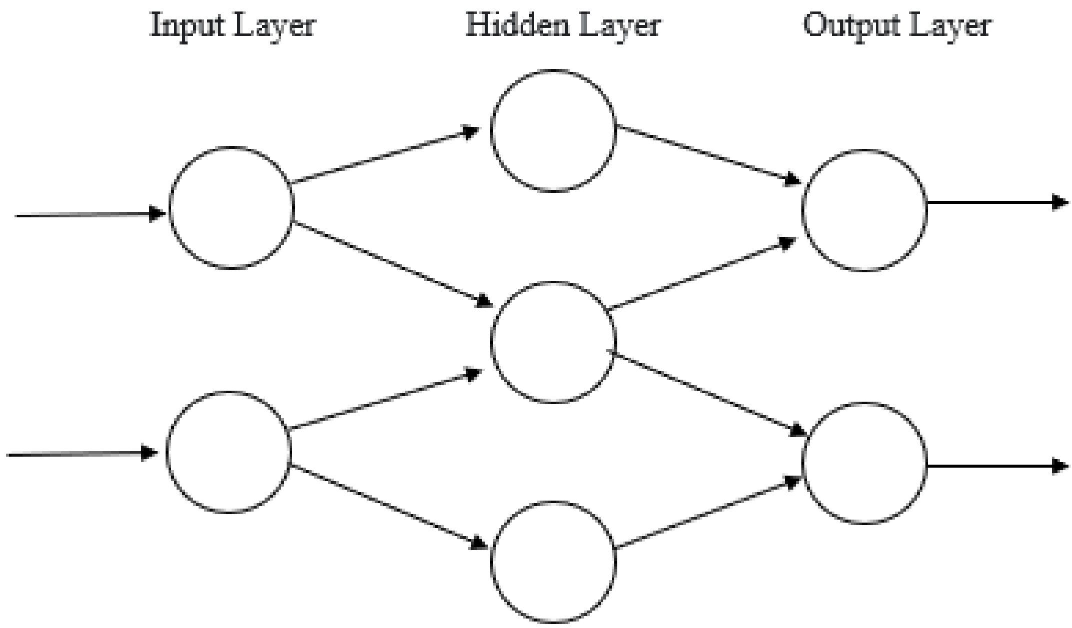


Figure 4.2: Block Diagram of ANN Model

4.5 Convolutional Neural Network

CNN is a type of machine learning algorithm where we provide enough input variables in the form of training, so that network can learn itself based on training input and gives accurate output. The layers of CNN contain the Input layer, a hidden layer and an output layer. It also includes multiple convolution layers, pooling layers. Figure 4.3 shows a block diagram of CNN Architecture.

The convolution network is a 2-D convolution layer. The CNN architecture is divided into two parts, Feature extraction and classification. In Feature extraction, 2-D input is convolved with filters and learns new features. These filters are called “weighted filters” . The max-pooling layer is used to reduce the size of convolution layers. Average pooling is also called Noise Suppressant. For the activation Function, ReLU (Rectified Linear Unit) performs the threshold operation. If the value of the weight filter is less than zero is set to zero. The experiment was performed on different learning rates and chose 0.01.

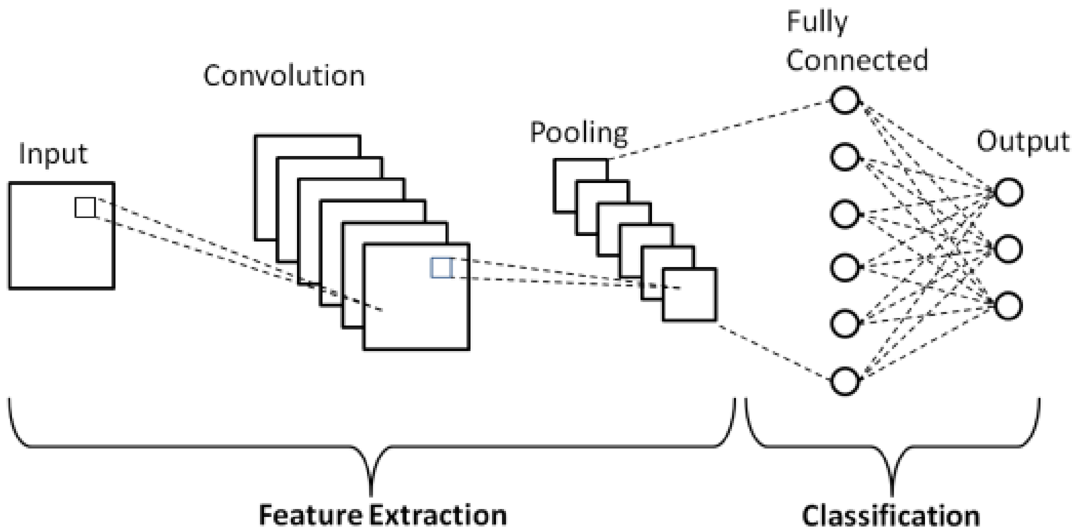


Figure 4.3: Block Diagram of CNN Architecture

The flatten layer converts all filter layers to the 1-D array. It is also called the Fully Connected layer. Each perceptron of layer learn feature and update weights. Finally, the softmax layer is used to calculate the probability of each species.

4.6 Summary

In this section, We have discussed different types of classification algorithms. For example, a deep learning approach called the CNN model contains two convolution layers and two flatten layers.

Chapter 5

Experimental Data and Results

5.1 Introduction

This section includes all the results achieved by using different feature extraction methods with various classifiers.The experiment is performed to test the performance of MFCC, LFCC and GFCC feature vector. The feature selection method with 27 coefficients. Each feature vector train and tested with k Nearest Neighbour (KNN), Artificial Neural Network (ANN), Support Vector Machine(SVM) and Convolutional Neural Network(CNN).

5.2 Experimental setup

Each classifier has been executed using python 3 and Jupyter Notebooks IDE. The machine for execution has a 2.6 GHz Core i5 processor with 4 GB RAM.

5.3 Data Description

Table 6.1 shows the list of bird files and scientific names used in this paper. There are 2358 individual files from 10 species. Each audio file is split into a 3-second chunk file. The spectral subtraction algorithm is used to reduce the noise in each chunk.

5.4 Result

This section discusses the result of all feature vectors of MFCC, LFCC and GFCC. The experiment has performed on 13,26 and 39 coefficients. In the confusion matrix, the best result is shown in Green color and the worst result is shown in Red color.

Table 5.1: Description of Bird Species Dataset

Sr.No.	Bird species	Scientific name	Number of Files
1	Red-whiskered Bulbul(RB)	Pycnonotus jocosus	236
2	Great Tilt(GT)	Parus major	243
3	Willow Warbler(WW)	Phylloscopus trochilus	262
4	European Pied Flycatcher(EF)	Ficedula hypoleuca	264
5	Common Blackbird(CB)	Turdus merula	256
6	Common Redstart(CR)	Phoenicurus phoenicurus	188
7	Greenish Warbler(GW)	phylloscopus torchiloides viridanus	207
8	Blue Jay(BJ)	Cyanocitta cristata	276
9	Common Snipe(CS)	Gallinago gallinago	244
10	Whistling Duck(WD)	Denfrocygna viduta	182
			Total 2358

5.5 Result for Feature Extraction

This section discusses the result of all feature vectors of MFCC, LFCC and GFCC. The experiment is performed on 13,26 and 39 coefficients. In the confusion matrix, the best result is shown in Green color and the worst result shown in Red color.

5.5.1 Result for MFCC

Table 5.2: Recognition of result with MFCC (13) features

	Number of correctly recognized bird sound in the class										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	36	1	2	0	0	0	0	1	4	0	82
GT	3	45	0	1	0	0	2	0	0	0	88
WW	2	0	37	4	0	0	1	4	0	0	77
EF	0	0	9	47	1	0	0	1	0	0	81
CB	0	0	2	6	43	2	0	0	0	0	81
CR	1	0	5	0	0	19	0	0	0	0	76
GW	4	1	3	0	0	0	48	1	0	0	84
BJ	0	0	0	0	0	0	4	43	8	6	70
CS	1	0	0	0	0	0	0	0	39	1	95
WD	1	0	0	0	0	0	0	2	0	36	92
Rel.	75	96	64	81	98	90	87	83	76	84	
Overall Accuracy: 82.38%											

Table 5.3: Recognition of result with MFCC (26) features

	Number of correctly recognized bird sound in the class										% RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	57	1	0	0	0	0	0	5	2	3	85
GT	0	55	0	13	3	2	4	0	1	0	67
WW	6	1	58	2	0	3	3	0	1	3	71
EF	0	21	24	25	4	7	6	0	0	2	37
CB	0	3	0	2	67	3	0	0	4	0	80
CR	0	1	0	2	0	63	0	2	0	0	85
GW	2	0	1	0	2	0	54	0	0	0	83
BJ	1	0	2	0	1	2	4	61	6	0	81
CS	0	3	2	0	11	0	0	0	48	3	73
WD	0	1	0	2	0	1	0	5	2	35	76
% Rel.	86	64	67	54	76	78	76	84	75	76	
Overall Accuracy: 81.33%											

Table 5.4: Recognition of result with MFCC (39) features

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	48	1	3	0	0	0	0	1	3	0	81
GT	0	45	2	9	2	2	1	0	2	0	66
WW	4	2	53	7	1	0	3	4	0	0	62
EF	1	20	25	31	4	1	1	0	0	1	42
CB	1	1	1	2	65	0	2	1	8	0	82
CR	3	2	7	5	0	37	0	4	0	3	72
GW	0	0	4	2	1	0	61	1	0	0	88
BJ	3	0	0	0	1	1	2	78	5	0	82
CS	2	0	1	2	2	0	0	11	55	1	74
WD	0	3	1	5	1	1	0	1	2	42	82
Rel.	77	61	55	49	84	88	87	77	73	89	
Overall Accuracy: 82.21%											

5.5.2 Result for LFCC

Table 5.5: Recognition of result with LFCC (13) features

	Number of correctly recognized bird sound										Acc %
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	50	0	1	0	0	0	0	2	3	0	89
GT	1	38	0	1	2	0	0	0	0	0	90
WW	0	0	37	9	0	1	2	0	0	3	71
EF	0	2	1	53	0	0	0	0	0	0	95
CB	0	0	1	5	42	0	1	0	0	0	86
CR	0	0	3	2	0	23	0	2	0	0	77
GW	0	0	3	0	0	0	57	1	0	0	93
BJ	2	0	0	0	0	0	4	39	1	3	80
CS	4	0	0	0	0	0	0	8	34	0	74
WD	0	0	0	0	1	0	0	0	0	34	97
% Rel	88	95	80	76	93	96	89	75	89	85	
Overall Accuracy: 85.53 %											

Table 5.6: Recognition of result with LFCC (26) features

	Number of correctly recognized bird sound in the class										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	44	1	9	0	0	1	0	6	3	0	69
GT	1	52	6	3	3	0	2	0	1	0	73
WW	4	1	53	8	0	0	2	6	2	1	56
EF	0	13	25	36	5	3	0	0	0	5	50
CB	3	2	1	5	59	0	3	1	3	0	79
CR	2	3	8	3	0	48	0	1	0	0	81
GW	2	2	3	0	0	0	49	1	0	0	83
BJ	2	0	7	0	1	0	5	56	8	4	71
CS	5	1	1	0	1	0	0	2	72	0	82
WD	1	0	0	1	3	2	0	1	5	35	75
Rel.	69	69	47	64	82	89	80	76	77	78	
Overall Accuracy: 85.49%											

Table 5.7: Recognition of result with LFCC (39) features

	Number of correctly recognized bird sound in the class										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	49	0	7	1	1	1	0	3	2	2	73
GT	1	56	1	5	2	0	2	4	2	0	75
WW	1	2	46	13	0	0	1	1	1	3	56
EF	0	10	14	39	7	2	0	0	0	12	50
CB	2	2	2	9	51	0	1	3	7	0	71
CR	0	2	10	4	0	35	0	0	0	0	84
GW	1	4	4	1	0	0	49	1	0	0	84
BJ	3	0	2	0	1	0	3	68	8	1	79
CS	5	0	0	0	1	0	0	6	64	0	86
WD	0	1	1	1	3	2	0	1	6	42	72
Rel.	72	73	55	53	77	88	88	78	71	70	
Overall Accuracy: 85.88%											

5.5.3 Result for GFCC

Table 5.8: Recognition of result with GFCC (13) features

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	56	0	4	1	1	0	0	3	4	0	74
GT	1	46	4	7	11	2	0	0	0	1	64
WW	17	4	39	2	1	2	5	2	0	0	46
EF	0	13	27	34	1	0	6	0	0	1	52
CB	1	4	0	1	64	0	7	0	3	0	77
CR	0	2	7	1	0	34	1	4	0	1	71
GW	0	0	2	3	1	0	54	0	0	0	81
BJ	3	0	2	0	0	2	0	77	4	0	81
CS	5	0	3	0	5	0	0	7	58	0	78
WD	0	2	10	0	2	6	0	8	1	28	64
Rel.	67	65	40	69	74	74	74	76	83	90	
Overall Accuracy: 87.64%											

Table 5.9: Recognition of result with GFCC (26) features

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	58	0	2	0	0	0	0	9	6	0	75
GT	0	40	6	10	11	0	0	0	0	3	65
WW	10	5	51	0	1	2	6	5	0	0	57
EF	0	3	32	46	0	1	6	0	0	3	59
CB	1	2	0	3	66	0	5	2	4	0	80
CR	1	0	4	5	0	36	1	8	0	3	72
GW	0	0	1	0	0	0	47	1	0	0	81
BJ	4	0	0	0	1	1	1	61	7	0	70
CS	5	0	2	0	2	0	0	5	59	0	79
WD	1	4	2	0	2	2	1	9	0	33	69
Rel.	72	74	51	72	80	86	70	61	78	79	
Overall Accuracy: 87.44%											

Table 5.10: Recognition of result with GFCC (39) features

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	46	0	2	0	0	0	0	5	3	0	75
GT	0	41	4	14	9	1	0	0	0	2	65
WW	13	3	50	3	0	3	3	4	0	0	53
EF	0	7	22	48	2	0	5	0	0	0	62
CB	1	0	0	1	62	0	7	0	2	0	82
CR	0	0	10	3	0	37	0	3	0	3	72
GW	0	0	2	2	0	0	64	0	0	0	86
BJ	2	0	3	0	1	6	1	76	2	0	78
CS	5	0	5	0	3	0	0	7	54	0	79
WD	0	4	10	0	2	0	0	8	1	31	67
Rel.	69	75	46	68	78	79	80	74	87	86	
Overall Accuracy: 85.88%											

Table 5.11: Overall Accuracy of Feature Extraction

Sr No.	**Feature vector**	**MFCC**	**LFCC**	**GFCC**
1	13 coefficients	82.38	85.53	87.64
2	26 coefficients	81.33	85.49	87.44
3	39 coefficients	82.21	85.88	85.88

Table 5.2,5.3 and 5.4 shows result with MFCC Features. Table 5.5,5.6 and 5.7 shows result with LFCC Feature. Table 5.8,5.9 and 5.10 shows the result of the GFCC Feature. Table 5.11 shows the overall accuracy of the Feature Extraction method. LFCC with 39 coefficients gives maximum accuracy of 85.88%.From the above result, it is observed that Willow Warbler(WW) and European Pied Flycatcher(EF) give bad results. Common Blackbird(CB) and Common Redstart(CR) give the best result in the Feature extraction technique

5.6 Selection of Optimum parameters

The choice of an optimum number of parameters is always desirable for the best classification. The Recursive Feature Elimination (RFE) algorithm is used for the selection of parameters in this work. The ranking of features is based on the priority of variable weight. The recursive feature elimination algorithm selects the only best feature and removes all other features. It is closely related to the Support Vector Machine(SVM) algorithm and reduces the dimensions of variables.

Recursive Feature Elimination (RFE) is a feature selection method that selects features and removes the weakest features till the specific number of features is reached. This model required a specific number of features. This process is iterated until all features in the dataset are exhausted. It conducts feature selection in a backward elimination procedure. It is a greedy optimization algorithm. The RFE algorithm heavily depends on the selection of feature sets and ranking. It constructs a subsequent model and ranks the feature based on the elimination of the feature.

Table 5.12 shows the result of RFE algorithm. A 'True' indicatesss that ethe feature is selected by the algorithm and is marked in blue color. The remaining/unwanted 'False' features are shown in orange. Thus, 27 'True' features are selected for further processing . The ranking of all 27 features is the same for all features, i.e. MFCC, LFCC and GFCC.

Based on Table 5.1, the features were added recursively from 1 to 27, and the accuracy was computed. It is observed that after 27 features are added recursively, the addition of remaining features results in a drop of accuracy. The corresponding graph is shown in Figure 5.1.

Table 5.12 to Table 5.17 shows the result of MFCC, LFCC and GFCC with SVM and KNN algorithm for using 27 features.

Table 5.12: Feature Selection

Coefficient Number	Selected Feature	Coefficient Number	Selected Feature	Coefficient Number	Selected Feature
1	True	14	True	27	False
2	True	15	True	28	False
3	True	16	True	29	False
4	False	17	True	30	False
5	True	18	True	31	False
6	True	19	True	32	False
7	True	20	True	33	True
8	False	21	True	34	True
9	True	22	True	35	True
10	True	23	True	36	True
11	True	24	True	37	False
12	False	25	True	38	False
13	True	26	True	39	False

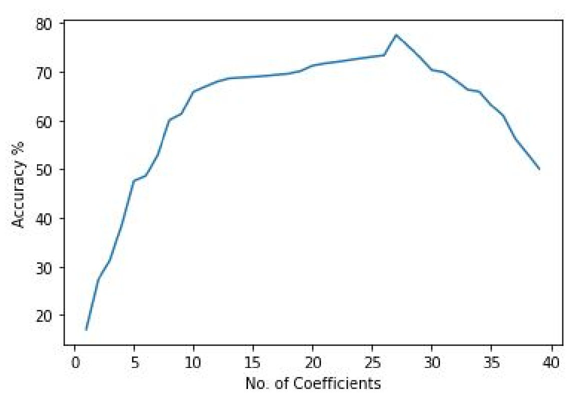


Figure 5.1: Accuracy vs number of features selected

5.6.1 Result for MFCC

Table 5.13: Recognition of result with MFCC (27) features

	Number of correctly recognized bird sound in the class										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	56	0	5	0	0	0	0	8	1	1	85
GT	0	61	0	2	3	2	1	0	1	1	74
WW	3	2	59	10	1	2	3	2	2	1	69
EF	0	21	21	30	2	2	0	0	0	0	45
CB	0	1	0	5	70	1	1	2	7	0	79
CR	0	3	0	1	0	43	0	1	0	1	81
GW	0	0	1	5	4	0	61	1	0	0	87
BJ	0	1	1	0	1	5	2	59	10	0	73
CS	1	4	0	1	9	0	0	6	44	0	65
WD	0	1	0	2	0	2	0	4	4	38	80
Rel.	93	65	68	54	78	75	90	71	64	86	
Overall Accuracy: 73.58%											

Table 5.14: Recognition of result with MFCC (27) features with KNN algorithm

	Number of correctly recognized bird sound in the class										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	72	1	0	0	0	0	0	2	0	1	96
GT	0	65	0	4	0	0	1	0	0	0	98
WW	0	0	61	12	0	1	0	0	0	0	82
EF	0	1	8	72	1	0	0	0	0	0	88
CB	0	0	0	7	77	1	1	0	0	0	90
CR	0	0	0	0	0	59	0	0	0	0	100
GW	0	0	0	0	0	0	69	1	0	0	100
BJ	0	0	0	0	0	0	3	75	0	0	100
CS	0	0	0	0	0	0	0	0	67	0	100
WD	0	0	0	0	0	0	0	1	0	54	98
Rel.	100	98	88	79	99	97	97	96	100	98	
Overall Accuracy: 94.70%											

5.6.2 Result of LFCC

Table 5.15: Recognition of result with LFCC (27) features

	Number of correctly recognized bird sound in the class										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	49	0	5	0	1	5	1	7	3	0	69
GT	0	57	5	5	1	0	1	1	0	1	80
WW	1	3	55	12	0	2	5	2	4	1	65
EF	0	1	18	35	5	1	0	4	0	0	46
CB	0	3	0	7	65	1	0	0	0	0	75
CR	0	1	0	0	0	55	0	0	0	0	80
GW	1	1	2	0	0	0	63	2	0	0	93
BJ	0	0	0	0	0	0	3	54	1	0	80
CS	2	0	0	0	0	0	0	0	54	0	91
WD	2	1	0	1	2	0	0	5	4	36	71
Rel.	88	71	61	57	89	75	85	69	68	77	
Overall Accuracy: 87.56%											

Table 5.16: Recognition of result with LFCC (27) features with KNN algorithm

	Number of correctly recognized bird sound in the class										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	66	1	0	0	0	0	0	0	0	0	99
GT	3	71	1	5	0	0	0	0	0	0	92
WW	0	0	58	13	0	1	0	0	0	0	79
EF	0	2	0	90	6	4	0	4	0	0	93
CB	0	0	0	5	72	0	0	0	0	0	97
CR	0	2	0	0	0	54	0	0	0	0	100
GW	0	2	0	0	0	0	54	0	0	0	96
BJ	0	0	0	0	0	0	0	78	1	0	99
CS	0	0	0	0	0	0	1	0	67	0	99
WD	0	0	0	0	0	0	0	0	0	63	100
Rel.	95	93	98	82	94	98	98	99	99	100	
Overall Accuracy: 95.05%											

5.6.3 Result of GFCC

Table 5.17: Recognition of result with GFCC (27) features

	Number of correctly recognized bird sound in the class										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	49	0	6	2	0	1	0	7	7	0	69
GT	0	48	3	7	7	0	0	0	0	6	68
WW	19	4	42	5	1	3	6	5	0	0	49
EF	0	5	24	41	2	0	0	0	0	0	54
CB	1	1	0	3	82	1	2	0	0	0	70
CR	1	0	5	3	0	30	0	3	0	7	61
GW	0	0	0	0	0	0	70	1	0	0	97
BJ	3	0	0	0	2	2	2	64	6	1	81
CS	8	0	2	3	4	0	0	7	43	0	64
WD	0	5	3	1	3	1	4	4	0	30	59
Rel.	60	76	49	62	77	83	69	71	69	68	
Overall Accuracy: 85.22%											

Table 5.18: Recognition of result with GFCC (27) features with KNN algorithm

	Number of correctly recognized bird sound in the class										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	62	0	0	0	0	0	0	0	0	0	100
GT	0	69	2	3	0	0	0	0	0	0	93
WW	0	2	57	10	0	1	0	0	0	0	83
EF	0	0	6	65	3	0	0	0	0	0	88
CB	1	0	0	5	87	0	2	0	0	0	94
CR	0	0	0	0	0	55	1	0	0	1	92
GW	0	0	0	0	0	0	61	0	0	0	98
BJ	0	0	0	0	0	0	3	73	1	0	94
CS	0	0	0	0	0	0	0	0	71	0	100
WD	1	3	7	3	0	2	0	0	0	49	75
Rel.	95	93	78	76	97	96	94	100	99	92	
Overall Accuracy: 91.66%											

Table 5.19: Overall Accuracy of Feature Selection

Sr. No.	Feature set	KNN	SVM
1	MFCC	94.7	73.58
2	LFCC	95.05	87.56
3	GFCC	91.66	85.22

Table 5.18 shows the overall accuracy of the feature selection technique with SVM and the KNN algorithm. LFCC with KNN algorithm gives an accuracy of 95.05 % From the above result, It is observed that European Pied Flycatcher(EF) gives bad results. Common Blackbird(CB) gives the best result in the Feature selection technique.

5.7 Classification

For classification, K-Nearest Neighbors (KNN) algorithm, Artificial Neural Network (ANN) algorithm and Convolutional Neural Network (CNN) algorithm are used.

5.7.1 Classification using KNN

5.7.1.1 Result of MFCC

Table 5.20: Recognition of result with MFCC (13) features with KNN algorithm

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	69	0	0	0	1	0	0	0	2	0	97
GT	0	66	1	8	0	0	0	0	0	3	91
WW	0	1	72	5	0	0	0	0	0	0	90
EF	0	0	9	68	0	0	0	0	0	0	84
CB	1	0	0	2	68	0	1	0	0	0	95
CR	0	0	0	2	0	54	1	0	0	1	96
GW	0	0	0	0	0	0	54	2	0	0	96
BJ	0	0	0	0	0	0	0	79	1	0	96
CS	0	0	0	0	2	0	0	0	74	0	97
WD	0	0	0	0	0	0	0	0	0	58	94
Rel.	99	99	88	80	96	100	96	94	96	94	
Overall Accuracy: 91.13%											

Table 5.21: Recognition of result with MFCC (26) features with KNN algorithm

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	62	1	0	0	0	0	0	0	0	0	99
GT	0	81	1	4	0	0	0	0	0	0	96
WW	0	1	61	11	0	0	0	0	0	0	84
EF	0	0	11	66	2	0	0	0	0	0	79
CB	0	0	0	6	71	1	1	0	1	0	93
CR	0	0	0	0	0	55	1	0	0	0	98
GW	0	0	0	0	0	0	69	0	0	0	98
BJ	0	0	0	0	0	0	1	80	2	0	98
CS	0	0	0	0	0	0	0	0	63	0	98
WD	0	0	0	0	1	0	0	1	0	54	
Rel.	100	98	84	75	97	98	96	99	95	1	
Overall Accuracy: 91.50%											

Table 5.22: Recognition of result with MFCC (39) features with KNN algorithm

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	68	2	0	0	5	0	0	1	0	0	94
GT	0	57	0	5	0	0	0	0	0	0	92
WW	0	1	65	12	0	0	0	0	0	0	87
EF	0	1	6	77	2	0	3	0	0	0	84
CB	0	0	0	1	72	0	1	0	0	0	94
CR	0	0	0	0	0	48	0	0	0	0	100
GW	0	0	0	0	0	0	60	1	0	0	96
BJ	0	1	1	0	0	0	0	85	0	0	97
CS	0	0	0	0	0	0	0	0	74	0	100
WD	1	0	0	0	0	0	0	1	0	57	98
Rel.	99	92	90	81	91	100	94	97	100	100	
Overall Accuracy: 93.64%											

5.7.1.2 Result of LFCC

Table 5.23: Recognition of result with LFCC (13) features with KNN algorithm

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	69	1	0	0	0	0	0	0	0	0	97
GT	3	65	1	0	0	0	0	0	0	0	91
WW	0	3	61	8	0	1	0	2	0	0	85
EF	0	0	5	61	6	4	0	4	0	0	79
CB	0	3	0	5	83	1	0	0	0	0	92
CR	0	1	0	0	0	55	0	0	0	0	94
GW	1	1	2	0	0	0	55	2	0	0	92
BJ	0	0	0	0	0	0	3	86	1	0	93
CS	0	0	0	0	0	0	0	0	63	0	99
WD	0	0	0	0	0	0	0	0	0	52	100
Rel.	95	88	88	82	93	90	95	91	98	100	
Overall Accuracy: 91.80%											

Table 5.24: Recognition of result with LFCC (26) features with KNN algorithm

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	69	2	0	0	0	0	0	0	0	0	96
GT	0	67	0	0	0	0	0	0	0	0	95
WW	0	1	67	7	0	0	0	2	0	0	85
EF	0	1	13	70	2	0	0	0	0	0	82
CB	1	0	0	6	73	0	0	0	1	0	94
CR	0	2	0	0	0	61	0	0	0	0	98
GW	0	0	0	0	0	0	60	2	0	0	96
BJ	0	0	0	0	0	0	3	78	0	0	96
CS	3	1	0	0	0	0	0	0	57	1	95
WD	0	0	0	1	0	0	0	0	0	57	98
Rel.	95	91	84	83	97	100	95	95	98	98	
Overall Accuracy: 91.76%											

Table 5.25: Recognition of result with LFCC (39) features with KNN algorithm

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	77	0	0	0	0	0	0	0	1	0	99
GT	0	63	1	0	0	0	0	0	0	0	98
WW	0	1	66	12	0	0	0	0	0	1	85
EF	0	1	4	70	3	0	0	0	0	0	85
CB	0	0	0	2	81	2	3	0	0	0	94
CR	0	0	2	2	0	49	0	0	0	0	94
GW	0	0	2	0	0	0	63	0	0	0	96
BJ	0	0	0	0	0	0	0	78	0	0	100
CS	1	0	0	0	0	0	0	0	69	2	97
WD	0	0	0	1	0	0	0	0	0	51	96
Rel.	99	97	88	80	96	96	95	100	99	94	
Overall Accuracy: 91.90%											

5.7.1.3 Result of GFCC

Table 5.26: Recognition of result with GFCC (13) features with KNN algorithm

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	65	0	0	0	0	1	0	2	2	0	95
GT	0	72	0	0	0	0	0	1	0	0	99
WW	0	1	70	13	0	1	0	0	0	0	88
EF	0	0	4	55	2	0	0	0	0	0	83
CB	1	0	0	4	82	1	2	0	0	0	94
CR	0	0	0	0	0	54	0	1	0	1	96
GW	0	0	0	0	0	0	66	1	0	0	97
BJ	0	0	0	0	0	0	1	80	1	1	94
CS	0	0	0	0	0	0	0	4	75	0	96
WD	1	0	0	0	0	0	0	0	0	44	97
Rel.	97	99	95	76	98	95	96	91	96	96	
Overall Accuracy: 91.64%											

Table 5.27: Recognition of result with GFCC (26) features with KNN algorithm

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	71	1	1	0	0	0	0	0	1	0	96
GT	0	72	1	0	0	0	0	0	0	0	97
WW	2	1	69	20	0	0	0	4	0	1	78
EF	0	2	7	72	5	0	0	0	0	0	79
CB	0	0	0	4	72	0	1	0	0	0	94
CR	0	0	0	0	0	45	0	0	0	0	100
GW	0	0	0	0	0	0	65	1	0	0	98
BJ	0	0	2	0	0	0	0	74	1	0	95
CS	0	0	0	0	0	0	0	0	70	0	99
WD	1	0	0	0	0	0	0	0	0	42	98
Rel.	96	95	86	75	94	100	98	94	97	98	
Overall Accuracy: 92.32%											

Table 5.28: Recognition of result with GFCC (39) features with KNN algorithm

	Number of correctly recognized bird sound										RSA
	RB	GT	WW	EF	CB	CR	GW	BJ	CS	WD	
RB	64	0	0	0	0	0	0	0	1	0	98
GT	0	71	0	2	0	2	0	0	0	0	95
WW	0	2	63	12	0	0	0	0	0	0	85
EF	0	1	4	81	3	0	0	0	0	0	86
CB	1	0	0	3	77	1	0	0	0	0	95
CR	0	0	0	0	0	60	0	0	0	1	97
GW	0	0	0	0	0	0	56	1	0	0	99
BJ	0	0	5	0	0	0	0	71	2	0	95
CS	0	0	0	0	0	0	0	0	75	0	98
WD	1	0	0	2	0	0	0	0	0	46	96
Rel.	97	96	88	81	96	95	100	99	96	98	
Overall Accuracy: 91.61%											

Table 5.29: Overall Accuracy of Feature Extraction using KNN algorithm

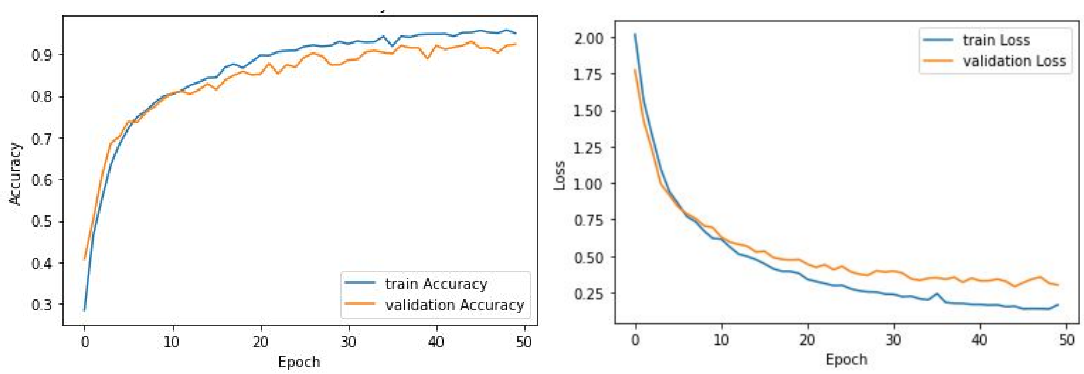
Sr No.	**Feature vector**	**MFCC**	**LFCC**	**GFCC**
1	13 coefficients	91.13	91.8	91.64
2	26 coefficients	91.5	91.76	92.32
3	39 coefficients	93.64	91.9	91.61

Table 5.28 shows the Overall accuracy of Feature Extraction using the KNN algorithm. MFCC with 39 coefficients gives the highest accuracy of 93.64 %. From the above result, it is observed that Willow Warbler(WW) and European Pied Flycatcher(EF) gives bad results. Common Blackbird(CB) gives the best result in the Feature extraction technique.

5.7.2 Classification using ANN

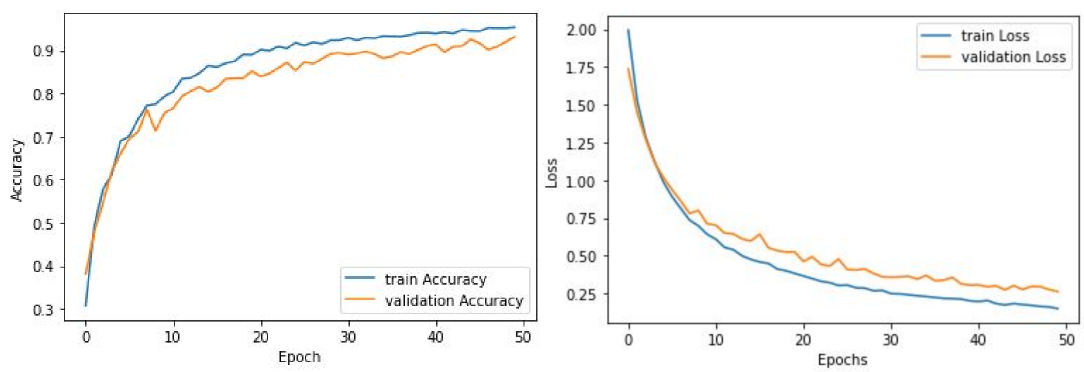
In the classification using the ANN algorithm, The graph shows Training and Validation accuracy and loss of MFCC, LFCC and GFCC features. I observed that at 20 epoch training accuracy achieved maximum accuracy.

5.7.2.1 Result of MFCC



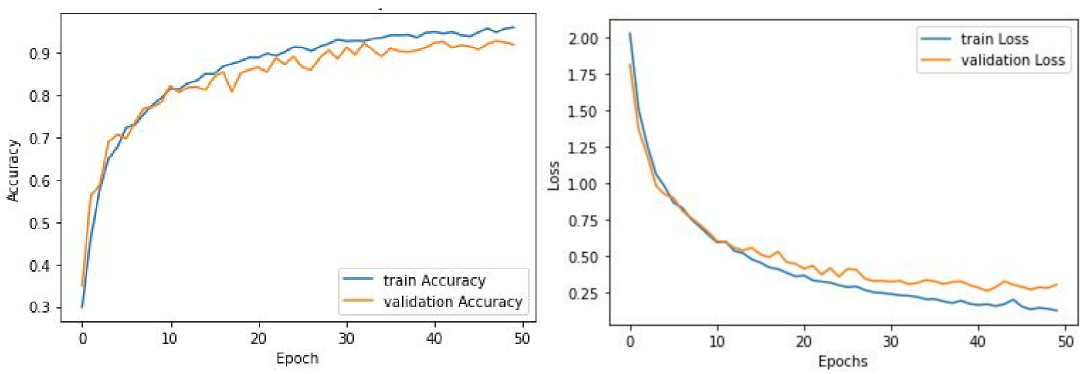
(a) (b)

Figure 5.2: Training and Validation Accuracy and loss of 13 MFCC coefficients



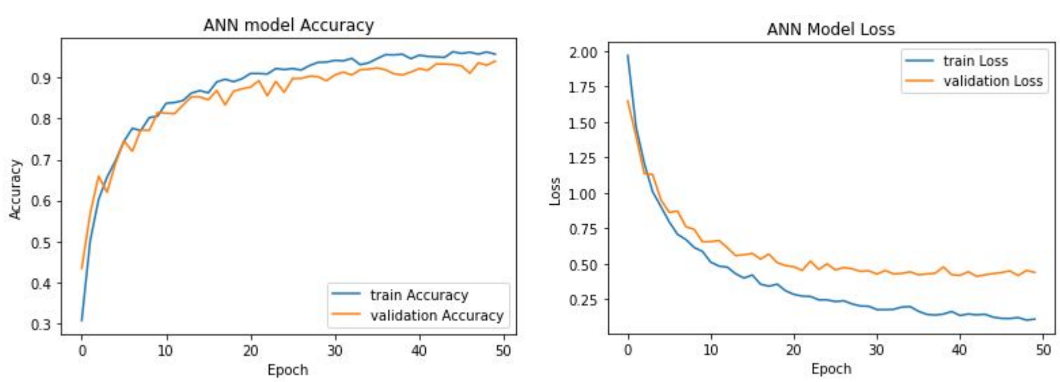
(a) (b)

Figure 5.3: Training and Validation Accuracy and loss of 26 MFCC coefficients



(a) (b)

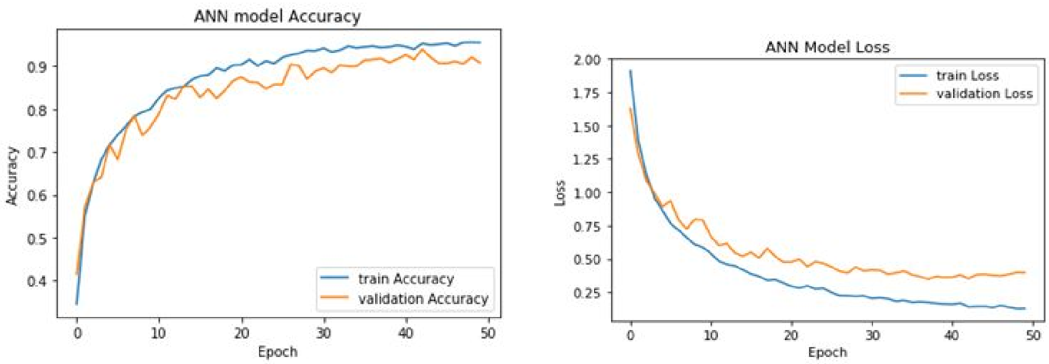
Figure 5.4: Training and Validation Accuracy and loss of 39 MFCC coefficients



(a) (b)

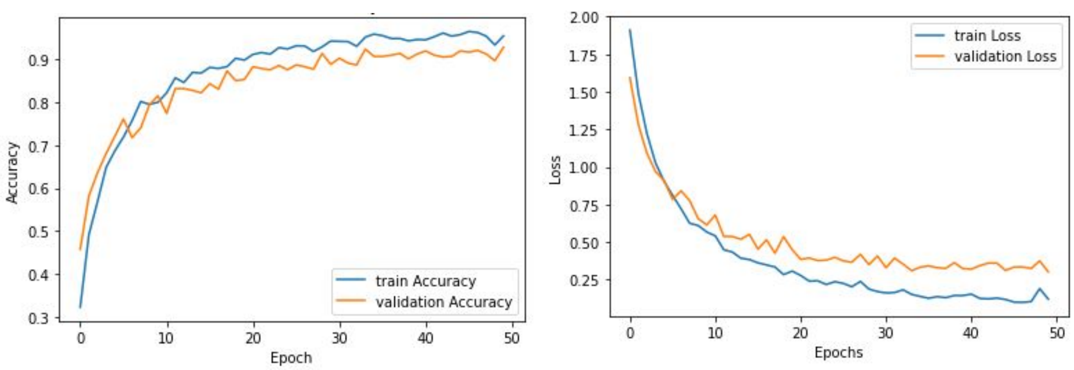
Figure 5.5: Training and Validation Accuracy and loss of 27 MFCC coefficients

5.7.2.2 Result of LFCC



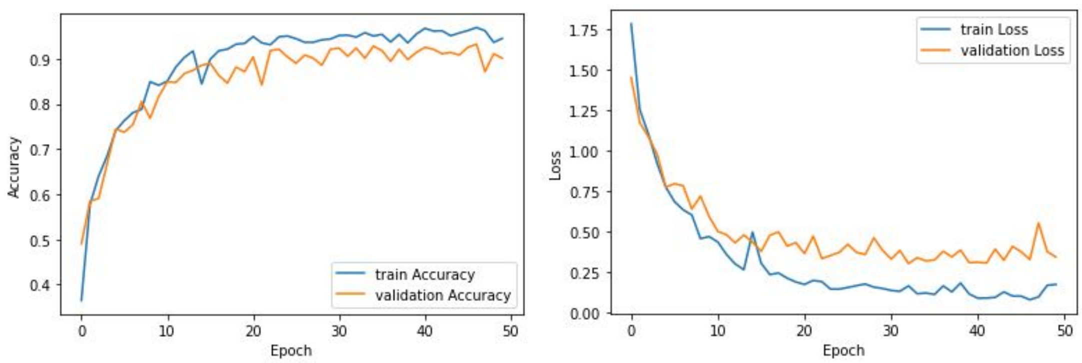
(a) (b)

Figure 5.6: Training and Validation Accuracy and loss of 13 LFCC coefficients



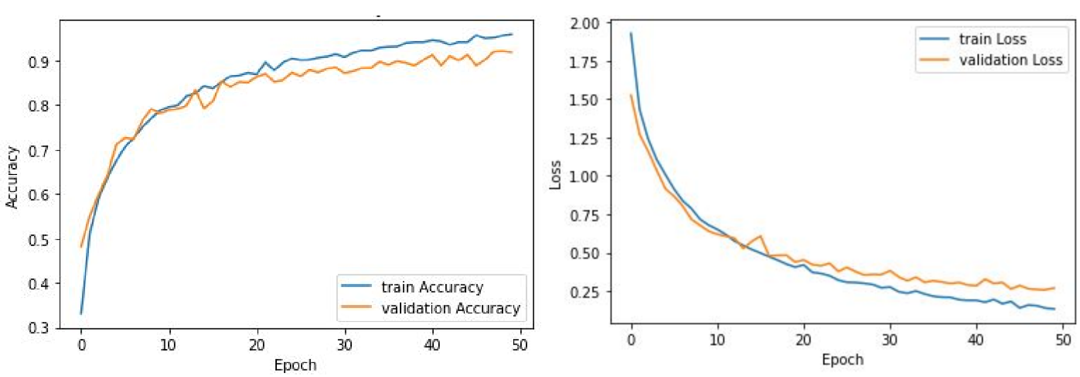
(a) (b)

Figure 5.7: Training and Validation Accuracy and loss of 26 LFCC coefficients



(a) (b)

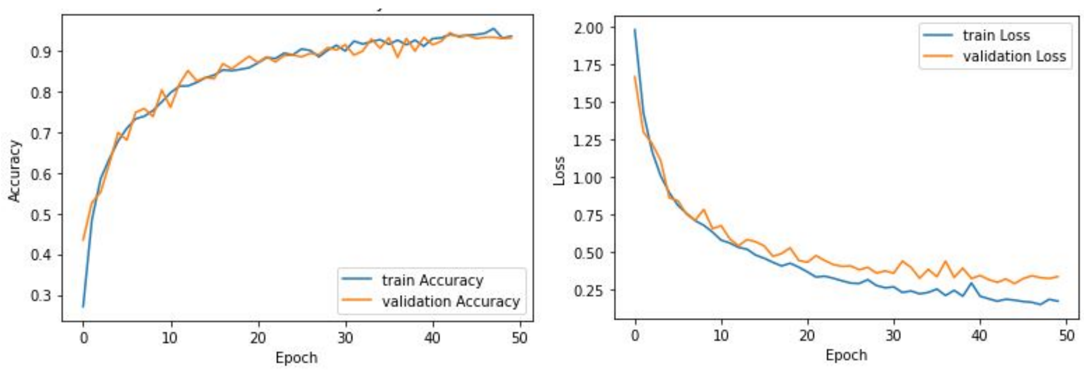
Figure 5.8: Training and Validation Accuracy and loss of 39 LFCC coefficients



(a) (b)

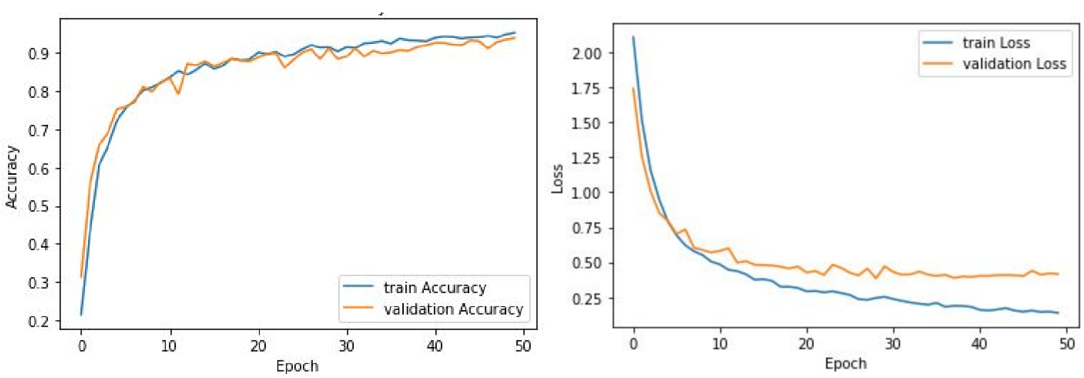
Figure 5.9: Training and Validation Accuracy and loss of 27 LFCC coefficients

5.7.2.3 Result of GFCC



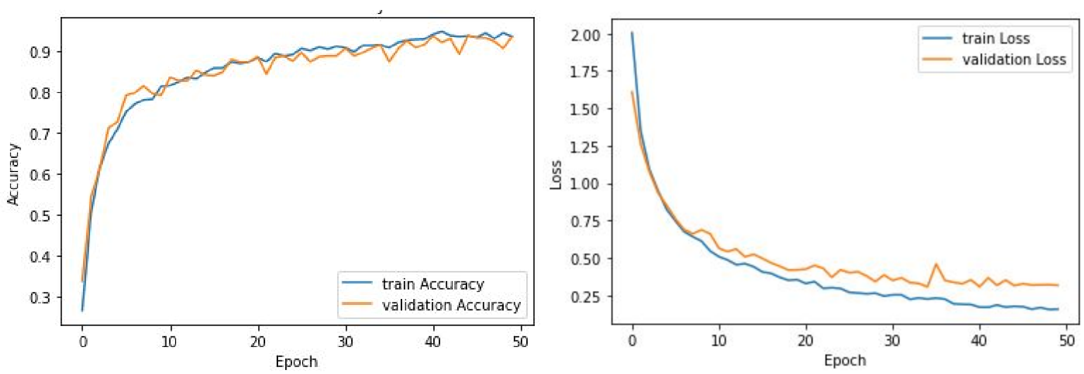
(a) (b)

Figure 5.10: Training and Validation Accuracy and loss of 13 GFCC coefficients



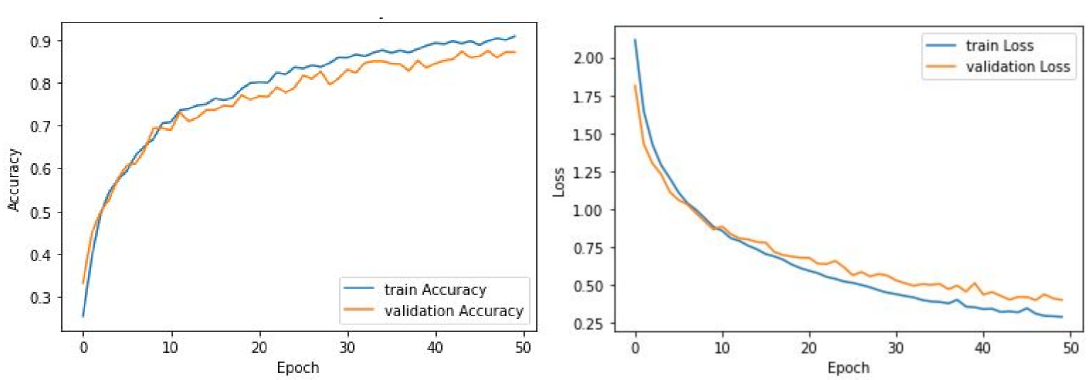
(a) (b)

Figure 5.11: Training and Validation Accuracy and loss of 26 GFCC coefficients



(a) (b)

Figure 5.12: Training and Validation Accuracy and loss of 39 GFCC coefficients



(a) (b)

Figure 5.13: Training and Validation Accuracy and loss of 27 GFCC coefficients

Table 5.30: Overall Accuracy of Feature Extraction using ANN algorithm

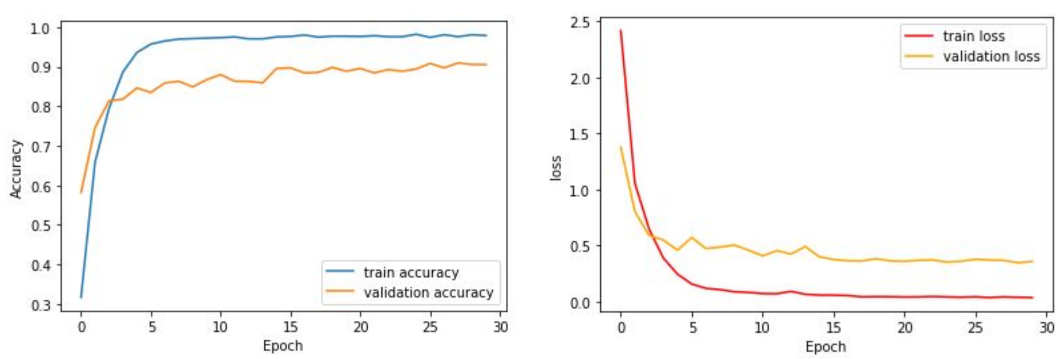
Sr No.	Feature vector	MFCC	LFCC	GFCC
1	13 coefficients	91.6	92.03	92.67
2	26 coefficients	92.07	91.66	92.87
3	39 coefficients	91.94	91.88	92.33
4	27 coefficients	94.7	92.6	87.88

Table 5.29 shows the overall accuracy of Feature extraction using ANN. MFCC with 27 coefficients gives maximum accuracy of 94.70 %. Figure 5.1,5.2,5.3 and 5.4 shows the Training and validation accuracy of MFCC coefficients. Figure 5.5,5.6,5.7 and 5.8 shows Training and validation accuracy of LFCC coefficients. Figure 5.9,5.10,5.11 and 5.12 shows the Training and validation accuracy of GFCC coefficients.

5.7.3 Classification using CNN Algorithm

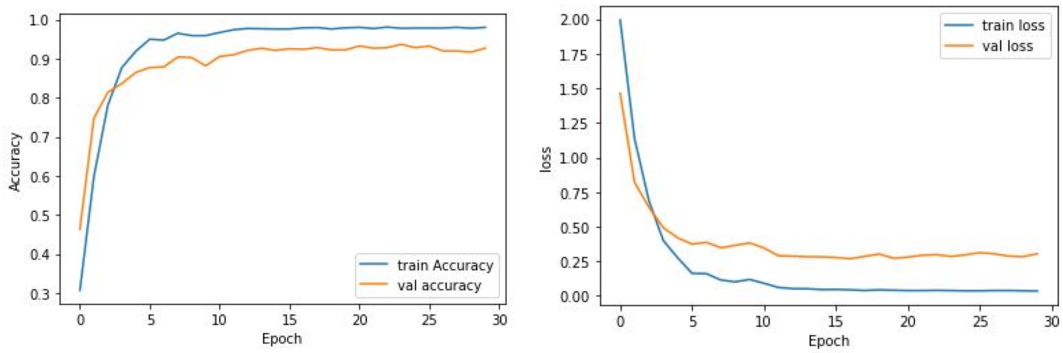
In the classification using the CNN algorithm, The graph shows Training and Validation accuracy and loss of MFCC, LFCC and GFCC features. I observed that at 10 epoch training accuracy achieved maximum accuracy.

5.7.3.1 Result of MFCC



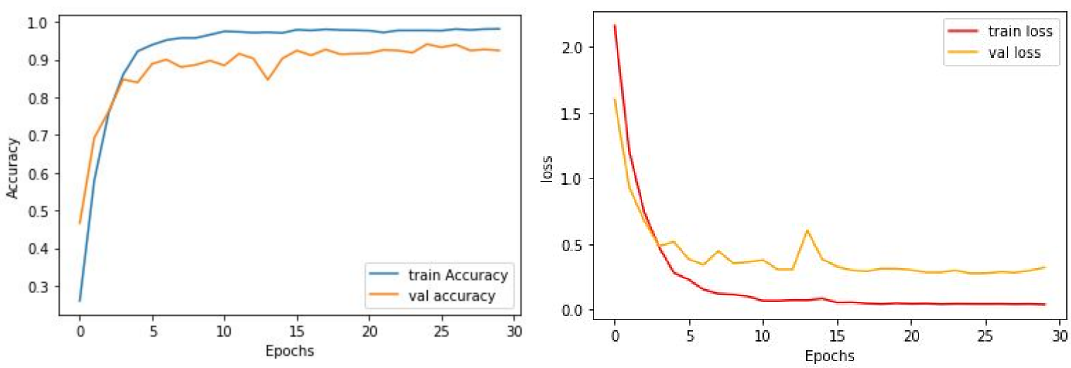
(a) (b)

Figure 5.14: Training and Validation Accuracy and loss of 13 MFCC coefficients



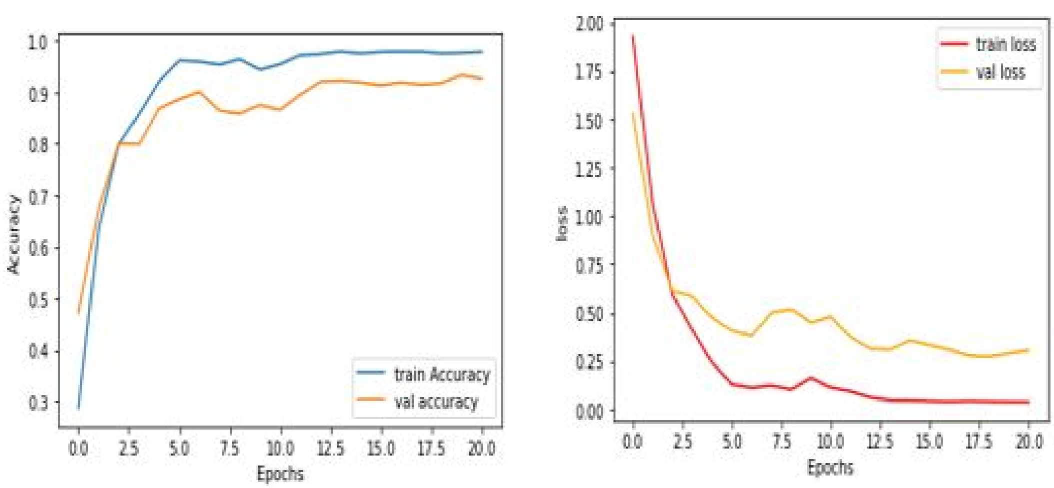
(a) (b)

Figure 5.15: Training and Validation Accuracy and loss of 26 MFCC coefficients



(a) (b)

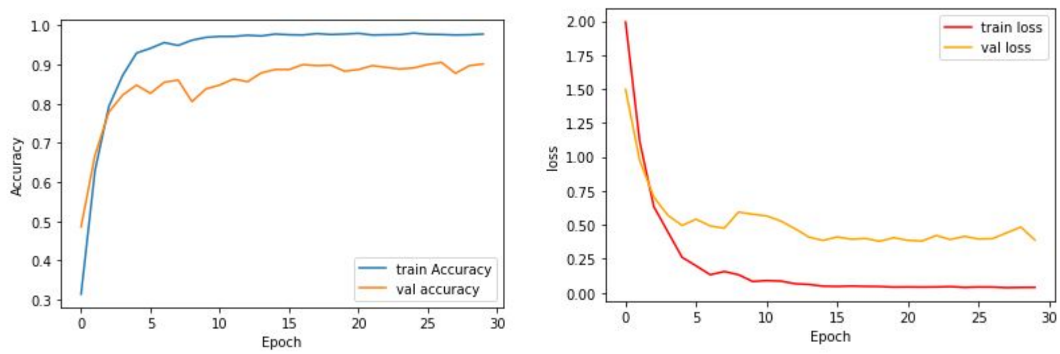
Figure 5.16: Training and Validation Accuracy and loss of 39 MFCC coefficients



(a) (b)

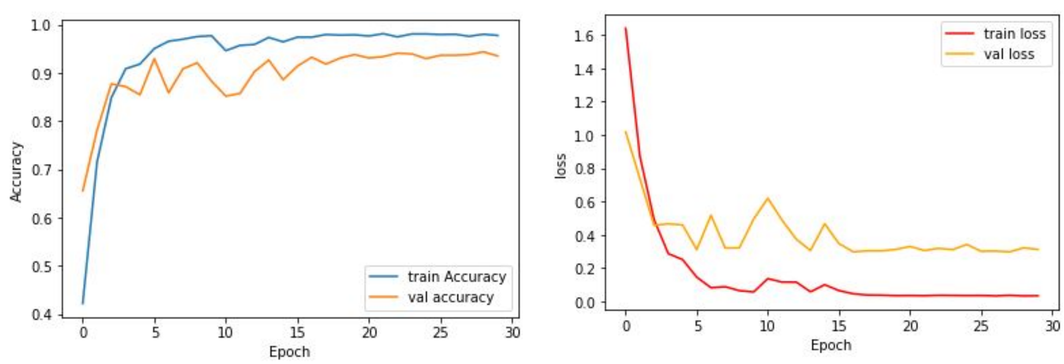
Figure 5.17: Training and Validation Accuracy and loss of 27 MFCC coefficients

5.7.3.2 Result of LFCC



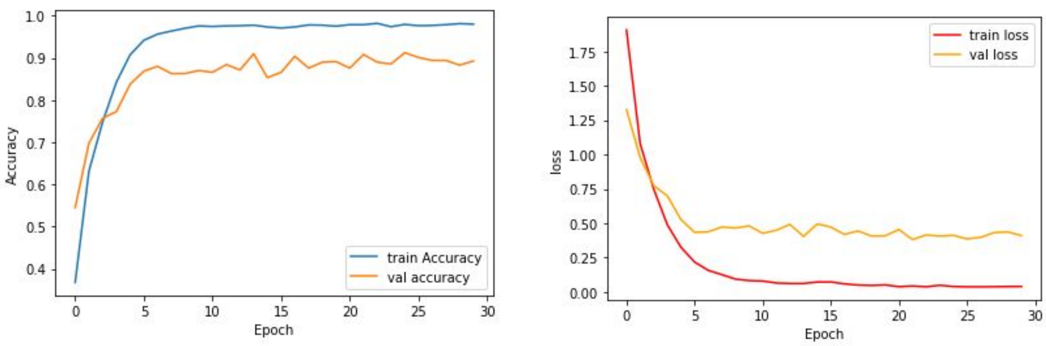
(a) (b)

Figure 5.18: Training and Validation Accuracy and loss of 13 LFCC coefficients



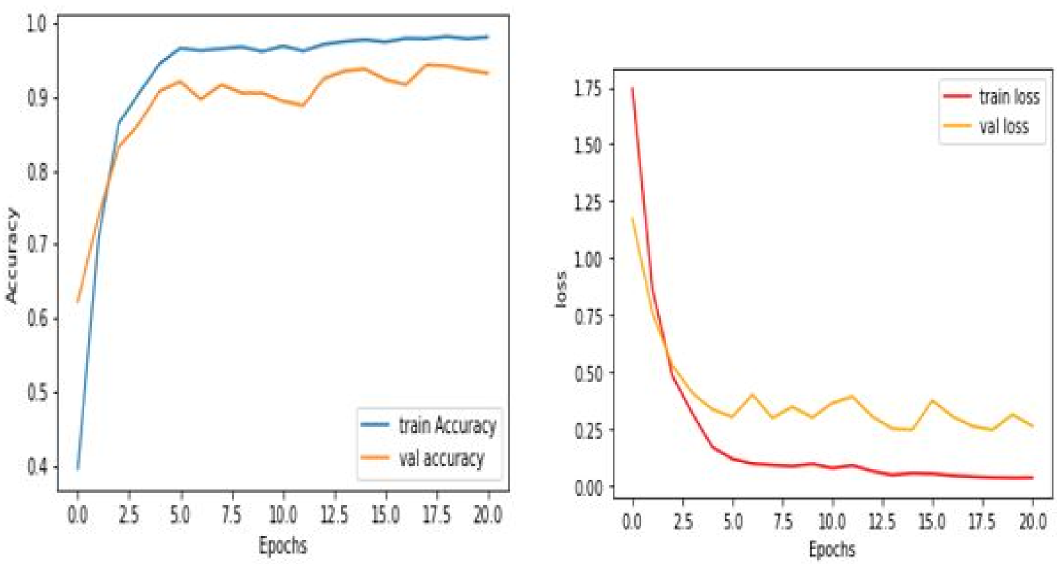
(a) (b)

Figure 5.19: Training and Validation Accuracy and loss of 26 LFCC coefficients



(a) (b)

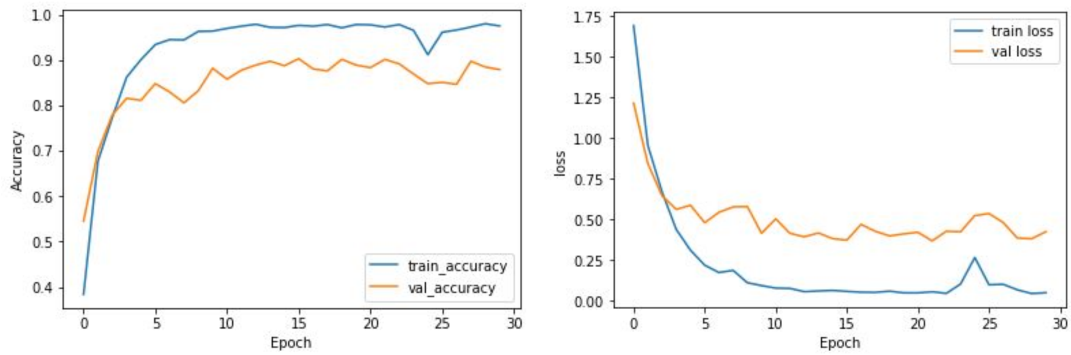
Figure 5.20: Training and Validation Accuracy and loss of 39 LFCC coefficients



(a) (b)

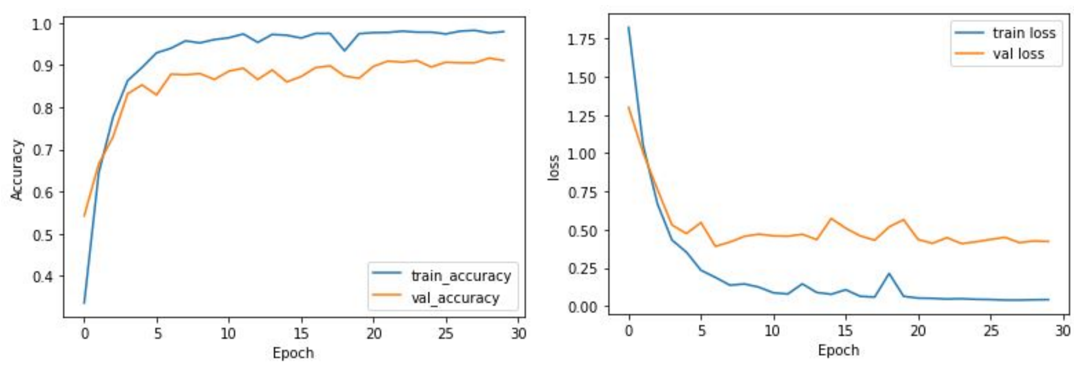
Figure 5.21: Training and Validation Accuracy and loss of 27 LFCC coefficients

5.7.3.3 Result of GFCC



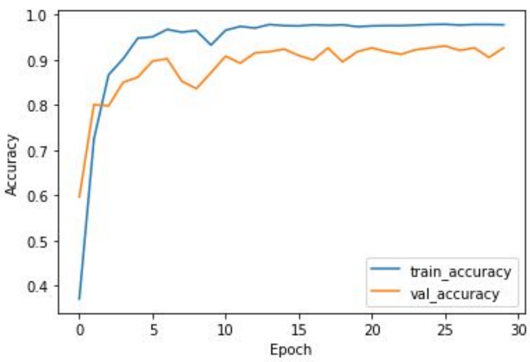
(a) (b)

Figure 5.22: Training and Validation Accuracy and loss of 13 GFCC coefficients

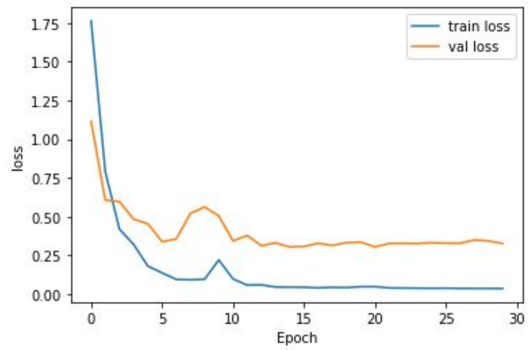


(a) (b)

Figure 5.23: Training and Validation Accuracy and loss of 26 GFCC coefficients

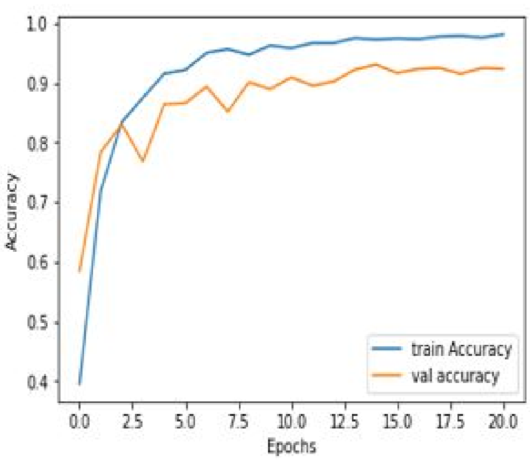


(a)

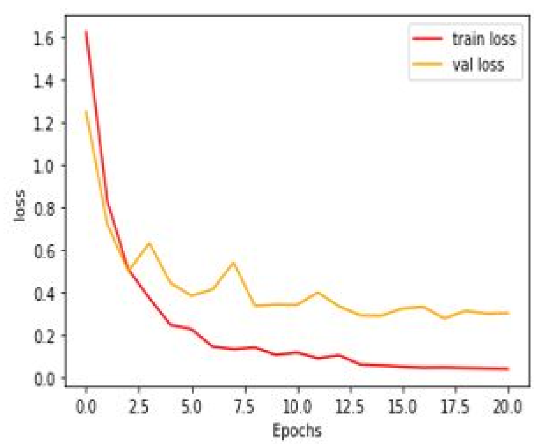


(b)

Figure 5.24: Training and Validation Accuracy and loss of 39 GFCC coefficients



(a)



(b)

Figure 5.25: Training and Validation Accuracy and loss of 27 GFCC coefficients

Table 5.31: Overall Accuracy of Feature Extraction using CNN algorithm

Sr No.	**Feature vector**	**MFCC**	**LFCC**	**GFCC**
1	13 coefficients	91.6	92.03	92.67
2	26 coefficients	92.07	91.66	92.87
3	39 coefficients	91.94	91.88	92.33
4	27 coefficients	94.7	92.6	87.88

Table 5.30 shows the Overall accuracy of Feature Extraction using the CNN algorithm.MFCC with 27 coefficients achieved maximum accuracy of 94.70 %

Table 5.32: Classification accuracy of Different Feature set and classifiers

Sr.No	Feature Set	Length of Feature Vector	Accuracy			
			KNN	SVM	ANN	CNN
1	Mel Frequency Cepstral Coefficients	13	91.13%	82.38%	91.60%	91.13%
		26	91.50%	81.33%	92.07%	90.36%
		39	93.64%	82.21%	91.94%	92.70%
		27	94.70%	73.58%	94.70%	93.60%
2	Linear Frequency Cepstral Coefficients	13	91.80%	85.53%	92.03%	91.67%
		26	91.76%	85.49%	91.66%	90.67%
		39	91.90%	85.88%	91.88%	92.80%
		27	95.05%	87.56%	92.60%	93.80%
3	Gammatone Frequency Cepstral coefficients	13	91.64%	87.64%	92.67%	91.58%
		26	92.32%	87.44%	92.87%	91.18%
		39	91.61%	85.88%	92.33%	90.18%
		27	91.66%	85.22%	87.88%	93.44%

Table 5.31 shows the summary of classification accuracy of different feature sets and classifiers.

5.8 Summary

The database of bird sounds was used for the classification of birds. The effect of features with various classifiers was explored in the chapter. It is observed that CNN gives the best result compared to other classifiers. The MFCCs outperform as features compared to LFCC and GFCC. The feature selection further improves the accuracy of classification.

Chapter 6

Conclusion and Future Scope

Three sets of classifiers and feature sets are selected to classify bird species. The testing accuracy of all feature set is evaluated. Gammatone Frequency Cepstral Coefficients with feature selection technique gives an accuracy of 93.44 %. Linear Frequency Cepstral Coefficients with feature selection techniques offers maximum accuracy of 95.05 %. The performance of the system is evaluated for different classifiers. The Recursive Feature Elimination (RFE) algorithm further improves the accuracy. CNN with MFCC as features gives the highest accuracy compared to LFCC and GFCC. Future work can be done by considering more feature selection methods with a different algorithm. Also, combining selected features with another algorithm will increase accuracy.

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